Capstone Project - Rail Equipment Accident/Incident Data (Form 54)

LINK:
https://data.transportation.gov/Railroads/Rail-Equipment-Accident-Incident-Data-Form-54-/85tf-25kj/about_data

Data Source:

- Title: Rail Equipment Accident/Incident Data (Form 54)
- Source: Department of Transportation - Open Source Portal [Last Updated: June 30th]
- Description: Data pulled for submitted "Form FRA 6180.54 - Rail Equipment Accident/Incident" (Form 54) to the FRA (Federal Railroad Administration). Forms are submitted 30 days after the last day of the month in which the incident occurred, detailing the initial incident and any additional outcomes between the incident date and the filing date.
- Rows: 224704 Records | Columns: 155 Features
- Timeframe: 1975 – 2026

Chosen Database:
Postgres SQL - more familiar with capabilities and able to handle large dataset

Context/background:
For my current company, I have been tasked with brainstorming a risk assessment model based on known characteristics for topic X. This project outlines the fundamental ideas/concepts I'll use. However, I'm focusing on a separate industry due to confidentiality. The railroad incident data was chosen for the capstone because it offers a similar layout to what I want: data collected in a form format, the form is completed after a set period of time, and it contains information about the current event and details from the day after.

Problem Statement:
Build both a predictive and a classification model for railway companies to use to predetermine the initial risk level and whether it will primarily be financial, operational, or legal after an incident occurs, preparing them with the right strategy before an investigation uncovers more details.

Hypothesis:
Are we able to predict the risk level associated with a railroad accident and whether it will point to a specific risk category using previously completed Form 54s, with only features that can be gathered at the time of the initial incident?

Methodology:

- Linear Regression: produce a "Risk Score"
- KNN: categorize the risk into Financial, Operational, or Legal "Risk Category"

Current Positon:

I have collected and briefly reviewed my data features. Implimented data into database as one massive data with no specific fix character limit attached. Will adjust later based on further exploration.

Next Steps & Experiment Plan:

1. Separate features into two groups: 1) Initial Features and 2) Outcome Features
2. Clean data
3. Exploratory data analysis
4. Build target variable from Group 2 ["Risk Score" and "Risk Category"]
5. "Risk Score" will be calculated via manually assigning weights to each feature in Group 2
6. "Risk Category" will be calculated via manually assigning if-then criteria for features in Group 2
7. Build model from Group 1 and target variable

Code:

I first uploaded that dataset to Python in Jupyter Notebook, since directly loading it into Postgres SQL was tedious. Using Claude AI and reviewing the data types, I discovered that some columns contained multiple data types. I took a different route and created the database in Jupyter Notebook with ease. The only downside is that all the features are set to one of the following (text, bigint, or double precision) without any specific character length. This will require me to do extensive data cleaning and then adjust my data schema to match accordingly

After the schema and data were stored in the database, I tested accessing the information in Jupyter Notebook by querying it. I then followed simple early data exploratory steps like .shape, .info(), and .describe(). For the time being, with the large number of missing values, I set every numeric feature to its mean. This also reflects how much data cleaning and generation will need to be completed.

I then attempted to generate both a distribution plot and a heatmap; however, due to the large number of features I am working with. The results came out poorly. Claude AI was used to attempt to resolve the issue, but no solution made the data any more visible. I decided to remove the heatmap portion and focus on five numeric features I thought would be strongly related to my data topic for the distribution plot.

In addition to complex problems, Claude AI and the text tools taught in the class lectures were used to troubleshoot coding mistakes, if any occurred, without a known solution.